AI-Based Resume Screening System Using Machine Learning and NLP

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Certificate

This is to certify that the project entitled **"AI-Based Resume Screening System Using Machine Learning and NLP"** has been carried out by **Team Firewall** under the guidance of **Subrat Nath Sharma .**

Project Guide
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Declaration

We hereby declare that the project report entitled **"AI-Based Resume Screening System Using Machine Learning and NLP"** is the result of our own work and has not been submitted to any other university or institution for the award of any degree or diploma.

Team Firewall

Acknowledgement

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Abstract

The **AI-Based Resume Screening System Using Machine Learning and NLP** automates the initial stage of recruitment by analyzing and shortlisting resumes intelligently. Traditional recruitment involves manual screening of numerous resumes, which is time-consuming and prone to human error. This project leverages **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)** to extract relevant information such as skills, qualifications, and experience from resumes in PDF format. Using machine learning algorithms, the system calculates a **match percentage** between the candidate's profile and the job requirements. The system aims to reduce HR workload, improve accuracy, and accelerate the hiring process. By integrating NLP for text extraction and analysis, the

system ensures efficient and unbiased candidate shortlisting, marking a significant step toward automation in recruitment.

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CHAPTER 1 — INTRODUCTION

1.1 Introduction to Resume Screening

Resume screening is the process of evaluating job applicants' resumes to identify candidates who best match the job requirements. It is the first and most crucial step in recruitment.

1.2 Traditional Recruitment Process

Traditionally, HR professionals manually review resumes, shortlist candidates, and schedule interviews. This process is labor-intensive and inefficient for large-scale recruitment.

1.3 Problems in Manual Resume Screening

Manual screening is prone to human bias, inconsistency, and fatigue. It consumes significant time and may overlook qualified candidates due to subjective judgment.

1.4 Need for Automation

Automation ensures faster, unbiased, and consistent evaluation of resumes. AI-based systems can process large datasets efficiently, improving recruitment quality.

1.5 AI in Recruitment

AI technologies, including machine learning and NLP, enable automated extraction of candidate information, skill matching, and predictive analysis for better hiring decisions.

CHAPTER 2 — LITERATURE REVIEW

2.1 Existing Resume Screening Methods

Conventional methods rely on keyword searches and manual filtering, which are inefficient for large applicant pools.

2.2 Applicant Tracking Systems (ATS)

ATS tools automate resume collection and keyword-based filtering but often fail to understand context or semantic meaning.

2.3 AI in HR Management

AI enhances HR operations by automating repetitive tasks, analyzing candidate data, and predicting job fit using data-driven insights.

2.4 NLP Applications in Recruitment

NLP enables machines to interpret human language, extract relevant information, and perform semantic analysis for skill and qualification detection.

CHAPTER 3 — PROBLEM STATEMENT

Recruiters face challenges in handling large volumes of resumes. Manual filtering is time-consuming, inconsistent, and error-prone. There is a need for an automated system that can efficiently analyze resumes, extract relevant data, and shortlist candidates based on predefined criteria.

CHAPTER 4 — OBJECTIVES

- Automate the resume screening process
- Reduce HR workload and time consumption
- Improve accuracy and consistency in candidate selection
- Enhance efficiency and fairness in recruitment

CHAPTER 5 — TECHNOLOGIES USED

5.1 Python

Python is used for its simplicity, extensive libraries, and strong support for machine learning and NLP.

5.2 Machine Learning

Machine learning algorithms analyze patterns in resumes and predict candidate suitability.

5.3 Natural Language Processing (NLP)

NLP processes textual data to extract skills, qualifications, and experience.

5.4 NLTK

The Natural Language Toolkit (NLTK) is used for tokenization, stemming, and text analysis.

5.5 Pandas

Pandas handles data manipulation and analysis efficiently.

5.6 PyPDF2

PyPDF2 extracts text from PDF resumes.

5.7 Google Colab

Google Colab provides a cloud-based environment for coding and testing.

5.8 Regular Expressions (re)

Regular expressions identify patterns such as email addresses, percentages, and qualifications.

CHAPTER 6 — SYSTEM REQUIREMENTS

6.1 Hardware Requirements

Component	**Specification**
Processor	Intel i5 or higher
RAM	8 GB minimum
Storage	500 GB HDD or SSD

6.2 Software Requirements

Software	**Version / Description**
Python	3.8 or above
Google Colab	Latest version
Libraries	NLTK, Pandas, PyPDF2, re

CHAPTER 7 — SYSTEM DESIGN

7.1 Architecture Diagram

(Placeholder for System Architecture Diagram)

7.2 Workflow Diagram

(Placeholder for Workflow Diagram)

Workflow:
Resume PDF Upload
↓
Text Extraction
↓
NLP Processing
↓
Skill Extraction
↓

Qualification Check
↓
Percentage Analysis
↓
Match Percentage Calculation
↓
Selected / Rejected Result

7.3 Data Flow Diagram

(Placeholder for Data Flow Diagram)

CHAPTER 8 — METHODOLOGY

1. **PDF Upload:** User uploads resume in PDF format.
2. **Text Extraction:** PyPDF2 extracts text content.
3. **Data Cleaning:** Remove unwanted symbols and stopwords.
4. **NLP Processing:** Tokenize and analyze text using NLTK.
5. **Skill Detection:** Match extracted words with predefined skill sets.
6. **Qualification Detection:** Identify educational qualifications.
7. **Percentage Extraction:** Extract academic percentages using regex.
8. **Match Calculation:** Compute match percentage based on job criteria.
9. **Final Prediction:** Display result as “Selected” or “Rejected”.

CHAPTER 9 — IMPLEMENTATION

9.1 Libraries Used

Library	Purpose
PyPDF2	Text extraction from PDF
NLTK	Text processing
Pandas	Data handling
re	Pattern matching

9.2 Code Explanation

The system reads resumes, extracts text, and applies NLP techniques to identify relevant information. Keyword matching and scoring logic determine the match percentage.

9.3 Logic Explanation

If the candidate's skills and qualifications meet the job requirements above a threshold (e.g., 70%), the candidate is shortlisted.

CHAPTER 10 — OUTPUT / RESULTS

10.1 Sample Output

Extracted Skills: Python, SQL
Qualification: B.Tech Graduate
Experience: Intern
Academic Percentage: 78%
Match Percentage: 85%
Final Result: Selected

Rejected Candidate Example:
Extracted Skills: HTML, CSS
Qualification: Diploma
Match Percentage: 45%
Final Result: Rejected

(Placeholder for Output Screenshots)

CHAPTER 11 — ADVANTAGES

- Saves time and effort
- Automates repetitive tasks
- Provides faster and unbiased results
- Reduces HR workload
- Improves recruitment efficiency

CHAPTER 12 — LIMITATIONS

- Dependent on keyword accuracy
- Limited understanding of context
- Formatting issues in resumes may affect extraction

CHAPTER 13 — FUTURE SCOPE

- Integration of deep learning models for semantic understanding
- Cloud-based deployment for scalability
- Integration with advanced ATS platforms
- AI-driven interview analysis

- Real-time recruitment dashboards

CHAPTER 14 — APPLICATIONS

- HR departments in organizations
- Recruitment agencies
- Job portals
- IT and corporate companies

CHAPTER 15 — CONCLUSION

The **AI-Based Resume Screening System** demonstrates the potential of AI and NLP in automating recruitment processes. By efficiently analyzing resumes and shortlisting candidates, the system reduces manual effort, enhances accuracy, and accelerates hiring. This project highlights the transformative role of AI in modern HR management and sets the foundation for future advancements in intelligent recruitment systems.

References / Bibliography

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Appendix

A. Source Code Snippets

(Placeholder for code snippets)

B. Screenshots

(Placeholder for screenshots of system interface and outputs)

End of Report